Capítulo 7

Lin 205C
Santiago Barreda

Chapter Precap

- Introducir modelos con varios factores con varios niveles.
- Discutir los conceptos de factores dentro y entre sujetos con respecto a los diseños factoriales, la ortogonalidad y las interacciones.
- Introducir la predicción a posteriori, y el uso de esta para la comprobación de modelos.
- Se discuten las interacciones y los gráficos de interacción, y como se utilizan para comprender los efectos principales y los efectos principales simples.
- Ajustamos un modelo con dos factores y una interacción, y efectos aleatorios para todos los predictores y el modelo se discute e interpreta.
- Presentamos el R^2 bayesiano como una medida simple del ajuste del modelo.
- Finalmente, discutimos los errores de tipo S y tipo M, las regiones de equivalencia práctica y el problema de cómo saber cuándo los efectos son "reales".

Comparing Many Groups

	A_1		A_2		A_3		A_4	
Within-Subjects	S_1	S_2	S_1	S_2	S_1	S_2	S_1	S_2
	S_3	S_4	S_3	S_4	S_3	S_4	S_3	S_4

	A_1		A_2		A_3		A_4	
Between-Subjects	S_1	S_2	S_5	S_6	S_9	S_{10}	S_{13}	S_{14}
	S_3	S_4	S_7	S_8	S_{11}	S_{12}	S_{15}	S_{16}

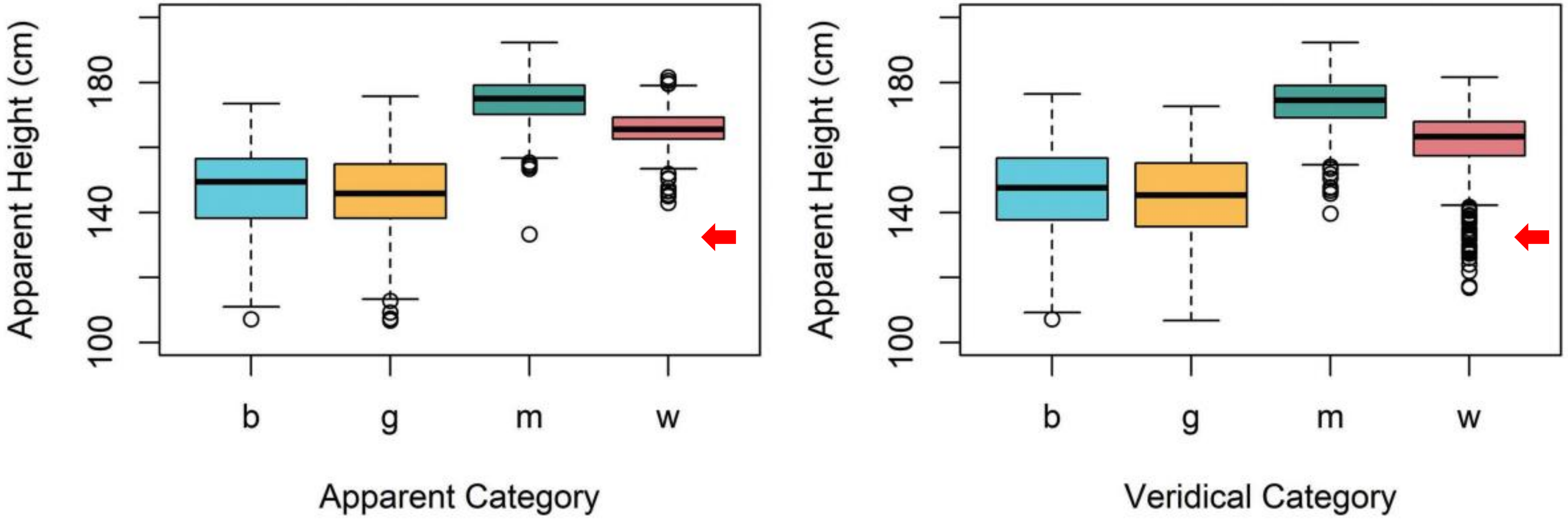
Data and Research Questions

```
library (brms)
library (bmmb)
options (contrasts = c('contr.sum','contr.sum'))
data (exp_data)
```

- L: A number from 1 to 15 indicating which *listener* responded to the trial.
- C: A letter representing the speaker *category* (b = boy, g = girl, m = man, w =woman) reported by the listener for each trial.
- height: A number representing the *height* (in centimeters) reported for the speaker on each trial.
- S: A number from 1 to 139 indicating which *speaker* produced the trial stimulus.

(Q1) Does apparent speaker height vary systematically across apparent speaker categories?

Apparent Speaker Category




```
# average correct category identification by category
tapply(exp_data$C == exp_data$C_v, exp_data$C_v, mean)
##      b      g      m      w
## 0.5778 0.6456 0.9274 0.7097
```

Comparing Many Groups

- Al estimar un factor con J niveles usando sum coding, solo puedes estimar (como máximo) valores únicos J-1.

- Los valores de los J niveles se restringen a la suma de cero con sum coding.

- Por lo tanto: El valor del nivel J es igual a la suma negativa de los niveles J-1.

$$A_1 + A_1 + A_3 + \cdots + A_{J-1} + A_J = 0$$

$$A_1 + A_1 + A_3 + \cdots + A_{J-1} = -A_J$$

$$-(A_1 + A_1 + A_3 + \cdots + A_{J-1}) = A_J$$

Description of the Model

4 niveles de C

```
height ~ C + (C|L) + (1|S)
```

Solo 3 medias de categoría predichas

```
height ~ C1 + C2 + C3 + (C1 + C2 + C3|L) + (1|S)
```

```
height ~ x1*C1 + x2*C2 + x3*C3 + (x1*C1 + x2*C2 + x3*C3|L) + (1|S)
```

No es una fórmula real

Description of the Model

```
height ~ x1*C1 + x2*C2 + x3*C3 + (x1*C1 + x2*C2 + x3*C3|L) + (1|S)
```

```
contr.sum(1:4)
##      [,1] [,2] [,3]
## 1       1    0    0
## 2       0    1    0
## 3       0    0    1
## 4      -1   -1   -1
```

$$\mathrm{C}_{[1]} = 1, \mathrm{C}_{[2]} = 2, \mathrm{C}_{[3]} = 3, \mathrm{C}_{[4]} = 4\ldots$$
$$\mu_{[i]} = x_1 \cdot C1 + x_2 \cdot C2 + x_3 \cdot C3$$

$$\mu_{[1]} = 1 \cdot C1 + 0 \cdot C2 + 0 \cdot C3 = C1$$
$$\mu_{[2]} = 0 \cdot C1 + 1 \cdot C2 + 0 \cdot C3 = C2$$
$$\mu_{[3]} = 0 \cdot C1 + 0 \cdot C2 + 1 \cdot C3 = C3$$
$$\mu_{[4]} = -1 \cdot C1 - 1 \cdot C2 - 1 \cdot C3 = -(C1 + C2 + C3) = C4$$

Description of the Model

$$\text{height}_{[i]} \sim \text{t}\left(\nu, \mu_{[i]}, \sigma\right)$$

$$\mu_{[i]} = \text{Intercept} + C_{\left[\mathsf{C}_{[i]}\right]} + L_{\left[\mathsf{L}_{[i]}\right]} + C_{\left[\mathsf{C}_{[i]}\right]} : L_{\left[\mathsf{L}_{[i]}\right]} + S_{\left[\mathsf{S}_{[i]}\right]}$$

$$\text{Priors :}$$

$$S_{[\cdot]} \sim \text{N}(0, \sigma_S)$$

$$\begin{bmatrix} L_{[\cdot]} \\ C_{[1]} : L_{[\cdot]} \\ C_{[2]} : L_{[\cdot]} \\ C_{[3]} : L_{[\cdot]} \end{bmatrix} \sim \text{MVNormal}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma\right)$$

$$\text{Intercept} \sim \text{N}(156, 12)$$

$$C_{[\cdot]} \sim \text{N}(0, 12)$$

$$\sigma_L, \sigma_{C[1]:L}, \sigma_{C[2]:L}, \sigma_{C[3]:L}, \sigma_S \sim \text{N}(0, 12)$$

$$\nu \sim \text{gamma}(2, 0.1)$$

$$R \sim \text{LKJCorr}(2)$$

Comparing Models

$$\text{height}_{[i]} \sim \text{t}\left(\nu, \mu_{[i]}, \sigma\right)$$
$$\mu_{[i]} = \text{Intercept} + C_{\left[\mathsf{C}_{[i]}\right]} + L_{\left[\mathsf{L}_{[i]}\right]} + C_{\left[\mathsf{C}_{[i]}\right]} : L_{\left[\mathsf{L}_{[i]}\right]} + S_{\left[\mathsf{S}_{[i]}\right]}$$

$$\text{Priors:}$$
$$S_{[\bullet]} \sim \text{N}(0, \sigma_S)$$
$$\begin{bmatrix} L_{[\bullet]} \\ C_{[1]} : L_{[\bullet]} \\ C_{[2]} : L_{[\bullet]} \\ C_{[3]} : L_{[\bullet]} \end{bmatrix} \sim \text{MVNormal}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma\right)$$

$$\text{Intercept} \sim \text{N}(156, 12)$$
$$C_{[\bullet]} \sim \text{N}(0, 12)$$
$$\sigma_L, \sigma_{C[1]:L}, \sigma_{C[2]:L}, \sigma_{C[3]:L}, \sigma_S \sim \text{N}(0, 12)$$
$$\nu \sim \text{gamma}(2, 0.1)$$
$$R \sim \text{LKJCorr}(2)$$

$$\text{height}_{[i]} \sim \text{t}\left(\nu, \mu_{[i]}, \sigma\right)$$
$$\mu_{[i]} = \text{Intercept} + A + L_{\left[\mathsf{L}_{[i]}\right]} + A : L_{\left[\mathsf{L}_{[i]}\right]} + S_{\left[\mathsf{S}_{[i]}\right]}$$

$$\text{Priors:}$$
$$S_{[\bullet]} \sim \text{N}(0, \sigma_S)$$
$$\begin{bmatrix} L_{[\bullet]} \\ A : L_{[\bullet]} \end{bmatrix} \sim \text{MVNormal}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma\right)$$

$$\text{Intercept} \sim \text{t}(3, 156, 12)$$
$$A \sim \text{t}(3, 0, 12)$$
$$\sigma, \sigma_L, \sigma_{A:L}, \sigma_S \sim \text{t}(3, 0, 12)$$
$$\nu \sim \text{gamma}(2, 0.1)$$
$$R \sim \text{LKJCorr}(2)$$

$$\Sigma = \begin{bmatrix} \sigma_L & 0 \\ 0 & \sigma_{A:L} \end{bmatrix} \cdot R \cdot \begin{bmatrix} \sigma_L & 0 \\ 0 & \sigma_{A:L} \end{bmatrix}$$

Fitting the Model

- ¡No hay priores nuevos!

```
# Fit the model yourself
priors = c(brms::set_prior("student_t(3,156, 12)", class = "Intercept"),
           brms::set_prior("student_t(3,0, 12)", class = "b"),
           brms::set_prior("student_t(3,0, 12)", class = "sd"),
           brms::set_prior("lkj_corr_cholesky (2)", class = "cor"),
           brms::set_prior("gamma(2, 0.1)", class = "nu"),
           brms::set_prior("student_t(3,0, 12)", class = "sigma"))

model_four_groups =
  brms::brm (height ~ C + (C|L) + (1|S), data = exp_data, chains = 4,
             cores = 4, warmup = 1000, iter = 5000, thin = 4,
             prior = priors, family = "student")

# or download it from the GitHub page:
model_four_groups = bmmb::get_model ('7_model_four_groups.RDS')
```

The Model Fixed Effects

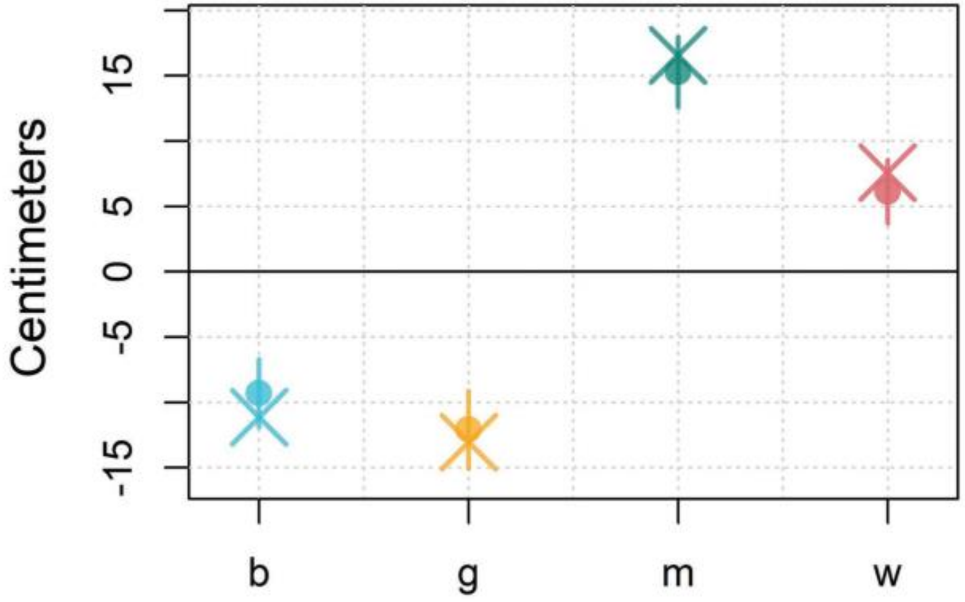
```
brms::fixef(model_four_groups)
##             Estimate Est.Error   Q2.5   Q97.5
## Intercept   158.581     1.083 156.41 160.685
## C1           -9.318     1.310 -11.87  -6.682
## C2          -12.082     1.494 -15.07  -9.110
## C3           15.286     1.327  12.59  17.950
```

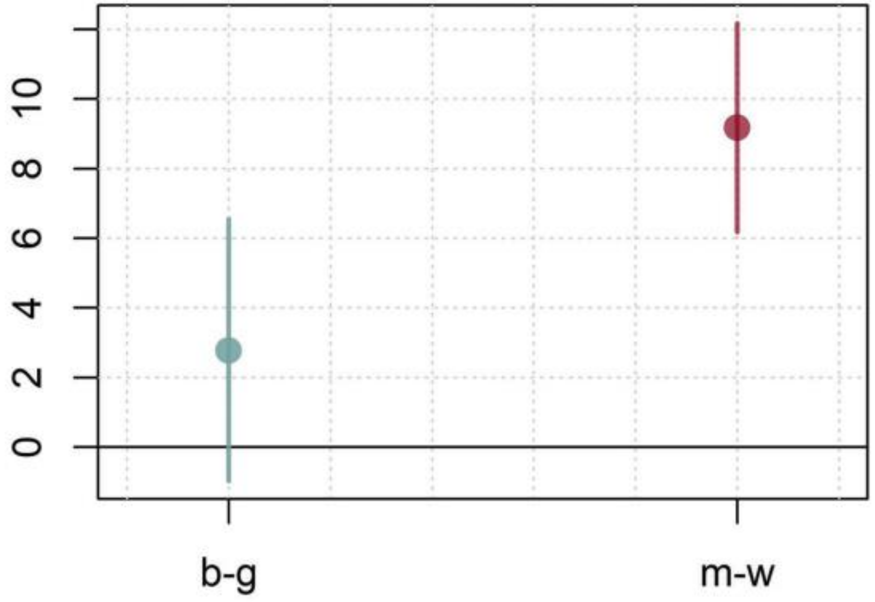
```
# group means
means = tapply (exp_data$height, exp_data$C, mean)
# Intercept = mean of means
mean (means)
## [1] 158

# Group effects = centered group means
means - mean (means)
##      b      g      m      w
## -11.12 -13.03  16.56   7.59
```

Working with the Fixed Effects

```
# missing group effect
bmmb::short_hypothesis (model_four_groups, c("-(C1+C2+C3) = 0"))
##     Estimate Est.Error  Q2.5 Q97.5         hypothesis
## H1     6.114     1.216 3.686 8.543 (-(C1+C2+C3)) = 0
```

```
# find differences between groups
comparisons = bmmb::short_hypothesis (model_four_groups,
                                      c("C1 = C2", "C3 = -(C1+C2+C3)"))
```





Multiple Factors at Once

Within-Subjects

	B_1	B_2
A_1	S_1 S_2 S_3 S_4	S_1 S_2 S_3 S_4
A_2	S_1 S_2 S_3 S_4	S_1 S_2 S_3 S_4

Between-Subjects

	B_1	B_2
A_1	S_1 S_2 S_3 S_4	S_5 S_6 S_7 S_8
A_2	S_9 S_{10} S_{11} S_{12}	S_{13} S_{14} S_{15} S_{16}

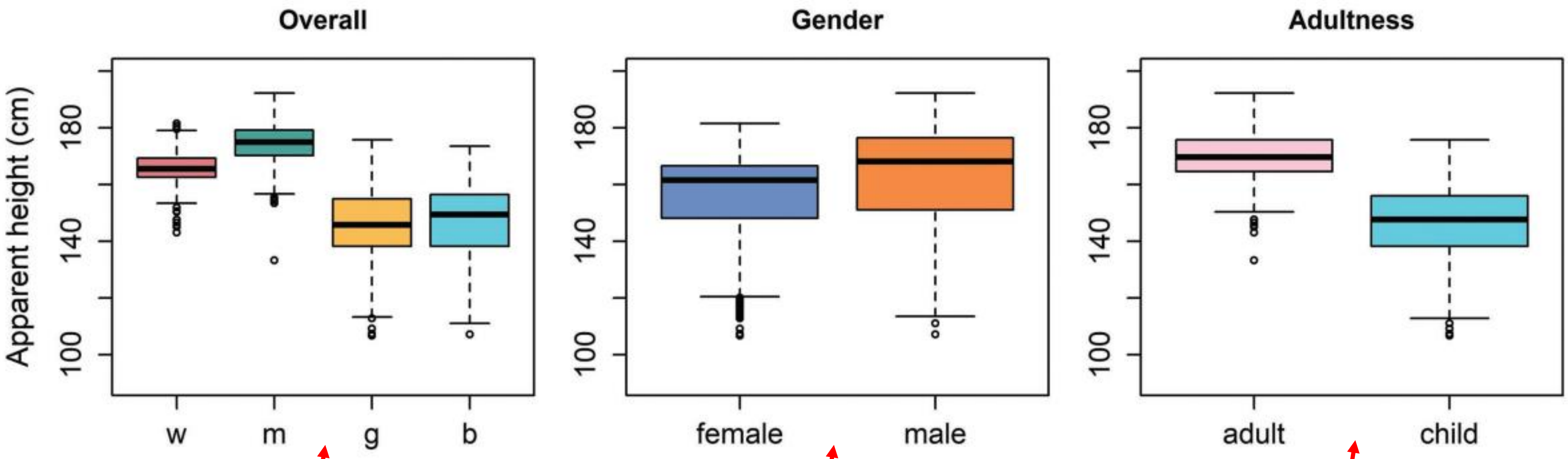
Mixed-design

	B_1	B_2
A_1	S_1 S_2 S_3 S_4	S_1 S_2 S_3 S_4
A_2	S_5 S_6 S_7 S_8	S_5 S_6 S_7 S_8

Diseño factorial/ortogonal

Una "célula" (es decir, una combinación de factores)

Multiple ~~Factors~~ Experiments at Once



¿Hay alguna diferencia?

¿Hay diferencias por género aparente?

¿Hay diferencias debido a la edad aparente?

Data and Research Questions

- `L`: A number from 1 to 15 indicating which *listener* responded to the trial.
- `height`: A number representing the *height* (in centimeters) reported for the speaker on each trial.
- `S`: A number from 1 to 139 indicating which *speaker* produced the trial stimulus.
- `G`: The *apparent gender* of the speaker indicated by the listener, `f` (female) or `m` (male).
- `A`: The *apparent age* of the speaker indicated by the listener, `a` (adult) or `c` (child).

(Q1) Does average apparent height differ across levels of apparent age?
(Q2) Does average apparent height differ across levels of apparent gender?

Description of the Model

~~brm (height ~ A + (A|L) + (1|S)~~

~~brm (height ~ G + (G|L) + (1|S)~~

```
height ~ A + G + (A + G|L) + (1|S)
```

$$\mu_{[i]} = \text{Intercept} + \left(C_{[\mathrm{C}[i]]} \right) \longrightarrow \mu_{[i]} = \text{Intercept} + \left(A_{[\mathrm{A}[i]]} + G_{[\mathrm{G}[i]]} \right)$$

Description of the Model

```
height ~ A + G + (A + G|L) + (1|S)
```

$$\text{height}_{[i]} \sim \mathrm{t}\left(\nu, \mu_{[i]}, \sigma\right)$$

$$\mu_{[i]} = \text{Intercept} + A + G + L_{\left[\mathsf{L}[i]\right]} + A : L_{\left[\mathsf{L}[i]\right]} + G : L_{\left[\mathsf{L}[i]\right]} + S_{\left[\mathsf{S}[i]\right]}$$

Priors:

$$S_{[\bullet]} \sim \mathrm{N}\left(0, \sigma_S\right)$$

$$\begin{bmatrix} L_{[\bullet]} \\ A : L_{[\bullet]} \\ G : L_{[\bullet]} \end{bmatrix} \sim \text{MVNormal}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma\right)$$

$$\text{Intercept} \sim \mathrm{N}(156, 12)$$

$$A, G \sim \mathrm{N}(0, 12)$$

$$\sigma_L, \sigma_{A:L}, \sigma_{G:L}, \sigma_S \sim \mathrm{N}(0, 12)$$

$$\sigma \sim \mathrm{N}(0, 12)$$

$$\nu \sim \text{gamma}(2, 0.1)$$

$$R \sim \text{LKJCorr}(2)$$

Fitting the Model

```
# Fit the model yourself
priors = c(brms::set_prior("student_t(3,156, 12)", class = "Intercept"),
           brms::set_prior("student_t(3,0, 12)", class = "b"),
           brms::set_prior("student_t(3,0, 12)", class = "sd"),
           brms::set_prior("lkj_corr_cholesky (2)", class = "cor"),
           brms::set_prior("gamma(2, 0.1)", class = "nu"),
           brms::set_prior("student_t(3,0, 12)", class = "sigma"))

model_both =
  brms::brm (height ~ A + G + (A + G|L) + (1|S), data = exp_data,
             chains = 4, cores = 4, warmup = 1000, iter = 5000,
             thin = 4, prior = priors, family = "student")

# Or download it from the GitHub page:
model_both = bmmb::get_model ('7_model_both.RDS')
```

Interpreting the Model

```
# inspect the fixed effects
brms::fixef (model_both)
##           Estimate Est.Error    Q2.5   Q97.5
## Intercept  158.654    1.1947 156.319 161.025
## A1           9.979    1.2107   7.562  12.282
## G1          -2.205    0.5524  -3.288  -1.124
```

```
# Intercept
mean (tapply (exp_data$height, exp_data$C, mean))
## [1] 158
```

```
# Age effect
diff (tapply (exp_data$height, exp_data$A, mean) ) / -2
##      c
## 12.16

# Gender effect
diff (tapply (exp_data$height, exp_data$G, mean) ) / -2
##       m
## -3.166
```

Predicting Group Means

```
means_pred = bmmb::short_hypothesis (model_both,
                                     c("Intercept + -A1 + -G1 = 0", # boys
                                       "Intercept + -A1 +  G1 = 0", # girls
                                       "Intercept +  A1 + -G1 = 0", # men
                                       "Intercept +  A1 +  G1 = 0")) # women

means_pred
##    Estimate Est.Error  Q2.5 Q97.5               hypothesis
## H1    150.9    2.2410 146.6 155.3 (Intercept+-A1+-G1) = 0
## H2    146.5    2.3680 142.0 151.2  (Intercept+-A1+G1) = 0
## H3    170.8    1.2988 168.2 173.4  (Intercept+A1+-G1) = 0
## H4    166.4    0.6901 165.0 167.8   (Intercept+A1+G1) = 0
```

```
tapply (exp_data$height, exp_data$C, mean)
##     b     g     m     w
## 146.9 145.0 174.5 165.6
```

Posterior Prediction

- La distribución predictiva posterior es la distribución de datos posibles dados los valores de los parámetros y el modelo de probabilidad.

model estimates

$$\tilde{y}_{[i]} \sim \mathrm{N}\left(\mu_{[i]}, \sigma\right)$$

- En una comprobación predictiva posterior (posterior predictive check), se comparan los datos simulados ($\tilde{y}$) con los datos reales (y).

Using Models to Predict

$$y_{[i]} \sim \mathrm{N}\left(\mu_{[i]}, \sigma\right)$$

predictor lineal/
Valor esperado

$$\boxed{\mu_{[i]}} = \text{Intercept} + A_{\left[\mathsf{A}[i]\right]} + B_{\left[\mathsf{B}[i]\right]} + C_{\left[\mathsf{C}[i]\right]}$$

```
# linear predictor
y_lin_pred = fitted (model_both)

# posterior prediction
y_post_pred = predict (model_both)
```

¡La predicción posterior incluye el error!

Los datos que predice el modelo dada su estimación de μ y σ para una observación.

$$\boxed{\tilde{y}_{[i]}} \sim \mathrm{N}\left(\mu_{[i]}, \boxed{\sigma}\right)$$

Using Models to Predict

```
# linear predictions
head (y_lin_pred)
##       Estimate Est.Error  Q2.5 Q97.5
## [1,]     156.8     2.182 152.5 161.0
## [2,]     161.7     2.059 157.6 165.6
## [3,]     160.6     1.983 156.6 164.3
## [4,]     162.3     1.846 158.5 165.9
## [5,]     163.0     1.992 159.0 167.0
## [6,]     155.7     2.371 151.1 160.5
```

```
# posterior predictions
head (y_post_pred)
##       Estimate Est.Error  Q2.5 Q97.5
## [1,]     157.0     7.915 141.3 172.2
## [2,]     161.7     9.137 146.3 177.2
## [3,]     160.5     8.210 144.8 175.9
## [4,]     162.1     8.312 146.4 177.0
## [5,]     162.8     7.639 147.8 177.9
## [6,]     155.7     7.708 140.0 170.8
```

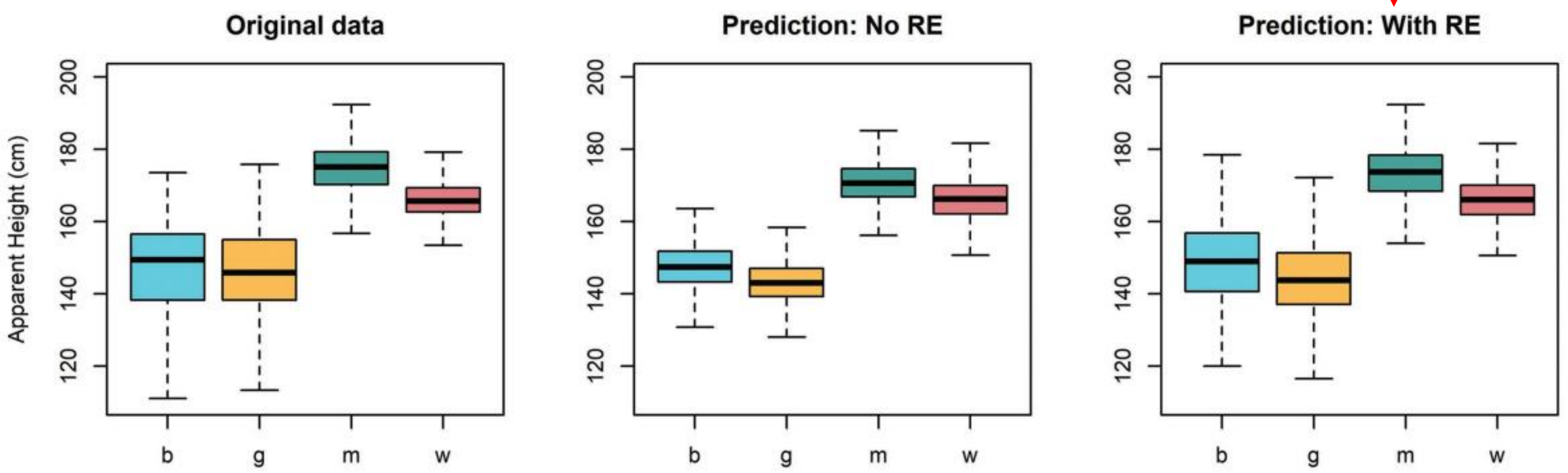
Fixed Effects Prediction

```
# posterior prediction
y_post_pred = predict (model_both)
```

height ~ A + G + (A + G|S) + (1|L)

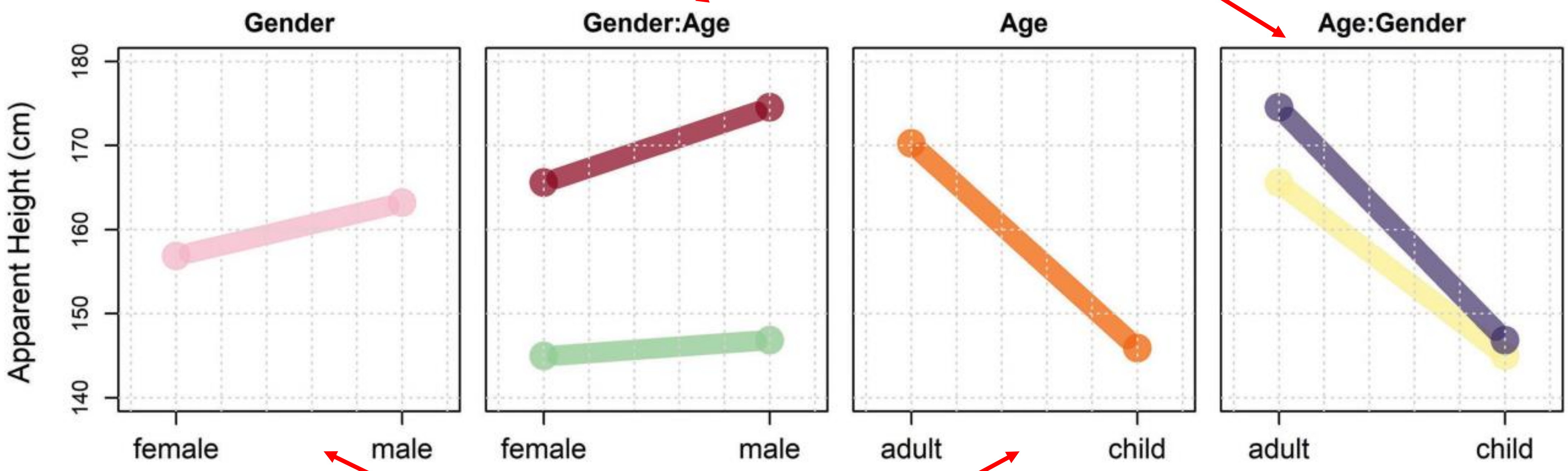
```
y_post_pred_no_re = predict (model_both, re_formula = NA)
```

height ~ A + G



Interaction Plots

Gráficos de interacción



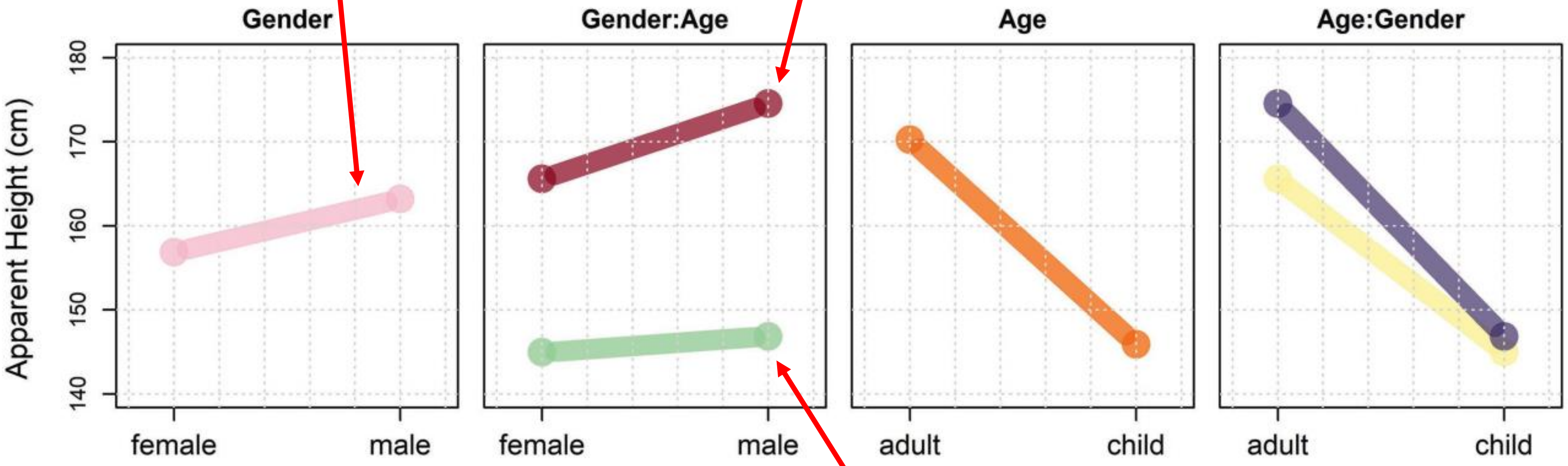
Gráficos de efectos principales

Simple (Main) Effects

- Efectos simples (principales): el efecto de un factor en un nivel de otro factor.

Efecto principal de género

Efecto simple de género dado un adulto aparente

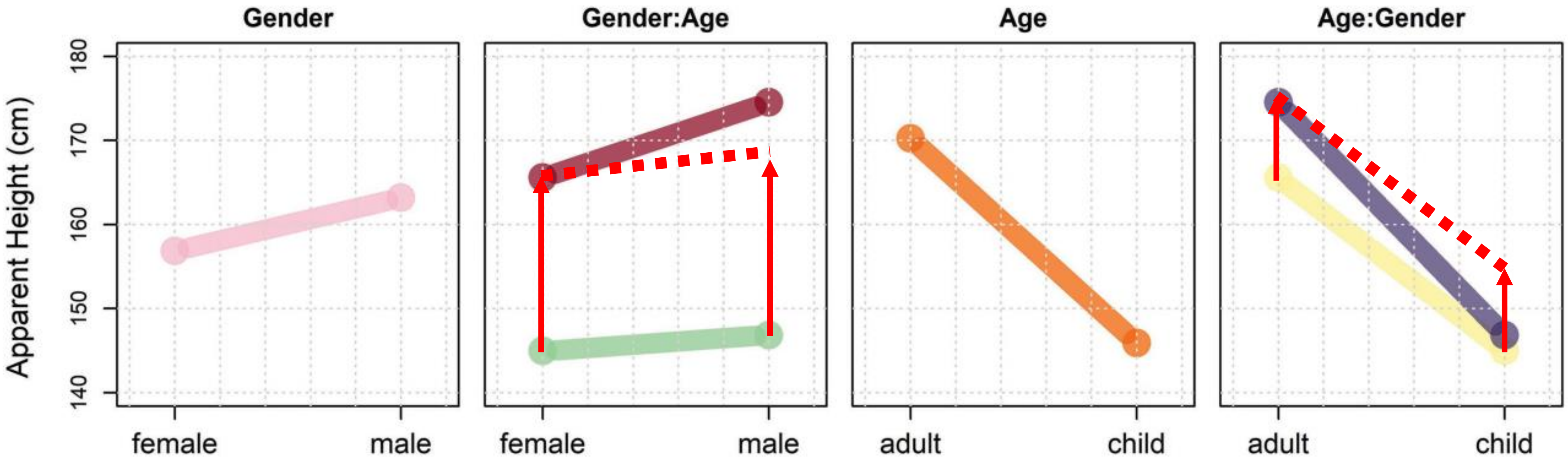


Efecto simple de género dado un niño aparente

Interaction Plots

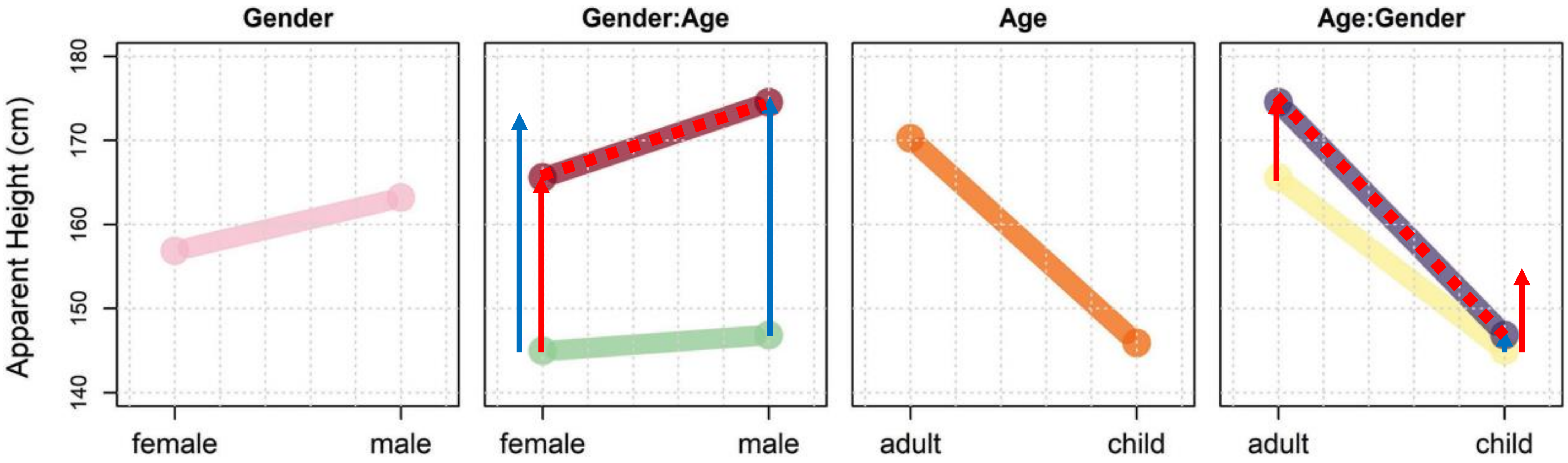
- En ausencia de una interacción, todos los puntos se desplazan en la misma cantidad (es decir, el efecto principal).

- Los efectos simples son paralelos.



Interaction Plots

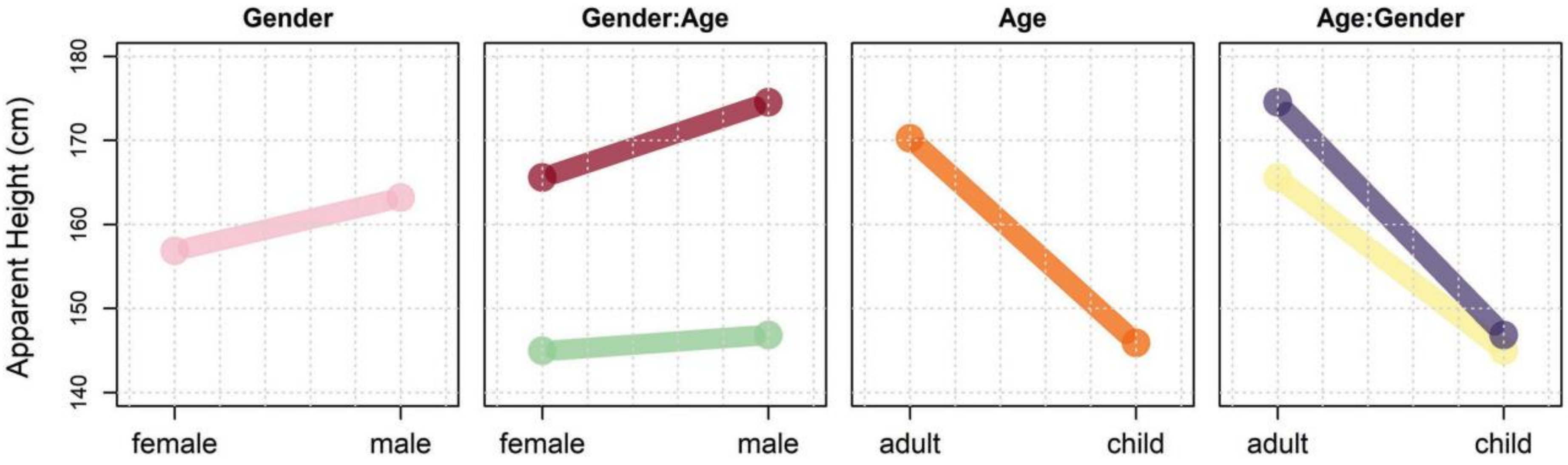
- Si existen interacciones, las líneas se pueden desplazar en diferentes cantidades en diferentes puntos (es decir, de acuerdo con los efectos condicionales).
- No paralelismo = ¡Interacción!



Simple (Main) Effects

- Cuando interacciones están presentes, los factores se interpretan en términos de sus efectos simples.

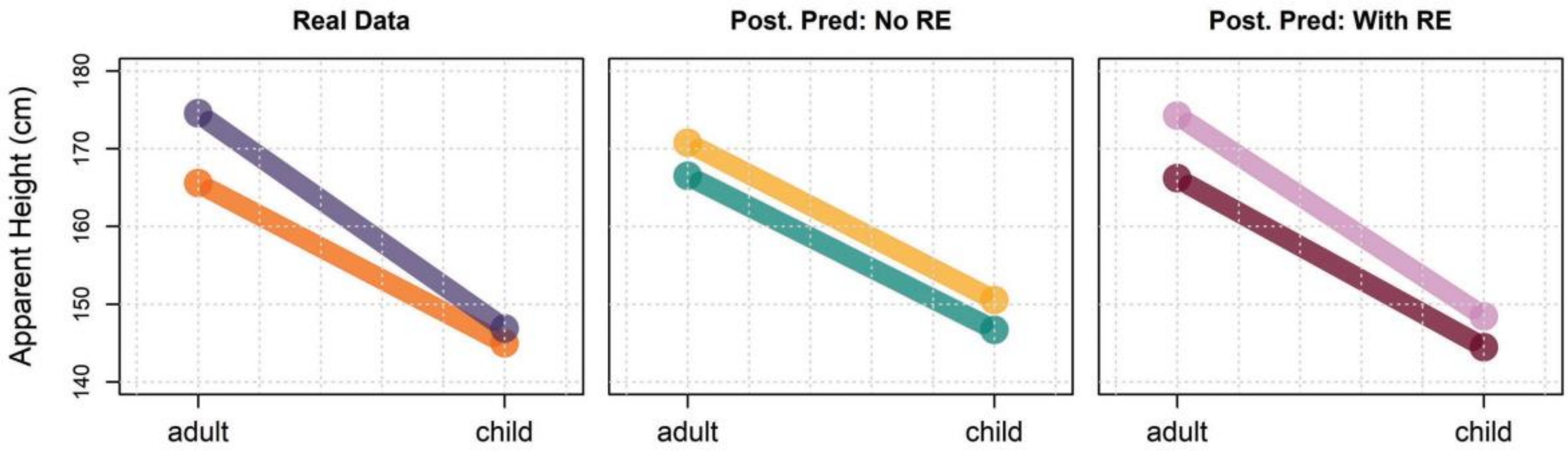
- Por ejemplo, a continuación: Sabemos que el efecto de género varía en función de la edad aparente. ¿Cómo?



No Interaction in Model = Bad

- No se puede saber si hay interacciones: los efectos simples tienen que ser iguales.

- Nuestros datos sugieren una interacción.



Description of the Model

```
height ~ A * G + (A * G|L) + (1|S)
```

```
height ~ A + G + A:G + (A + G + A:G|L) + (1|S)
```

$$
\begin{aligned}
\text{height}_{[i]} &\sim \mathrm{t}\left(\nu, \mu_{[i]}, \sigma\right) \\
\mu_{[i]} &= \text{Intercept} + A + G + A:G + \\
&L_{\left[\mathsf{L}[i]\right]} + A:L_{\left[\mathsf{L}[i]\right]} + G:L_{\left[\mathsf{L}[i]\right]} + A:G:L_{\left[\mathsf{L}[i]\right]} + S_{\left[\mathsf{S}[i]\right]}
\end{aligned}
$$

Priors :

$$
S_{[\bullet]} \sim \mathrm{t}(3, 0, \sigma_S)
$$

$$
\begin{bmatrix} L_{[\bullet]} \\ A:L_{[\bullet]} \\ G:L_{[\bullet]} \\ A:G:L_{[\bullet]} \end{bmatrix} \sim \text{MVNormal}\left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma \right)
$$

$$
\begin{aligned}
\text{Intercept} &\sim \mathrm{t}(3, 156, 12) \\
A, G, A:G &\sim \mathrm{t}(3, 0, 12) \\
\sigma_L, \sigma_{A:L}, \sigma_{G:L}, \sigma_{A:G:L}, \sigma_S &\sim \mathrm{t}(3, 0, 12) \\
\sigma &\sim \mathrm{t}(3, 0, 12) \\
\nu &\sim \text{gamma}(2, 0.1) \\
R &\sim \text{LKJCorr}(2)
\end{aligned}
$$

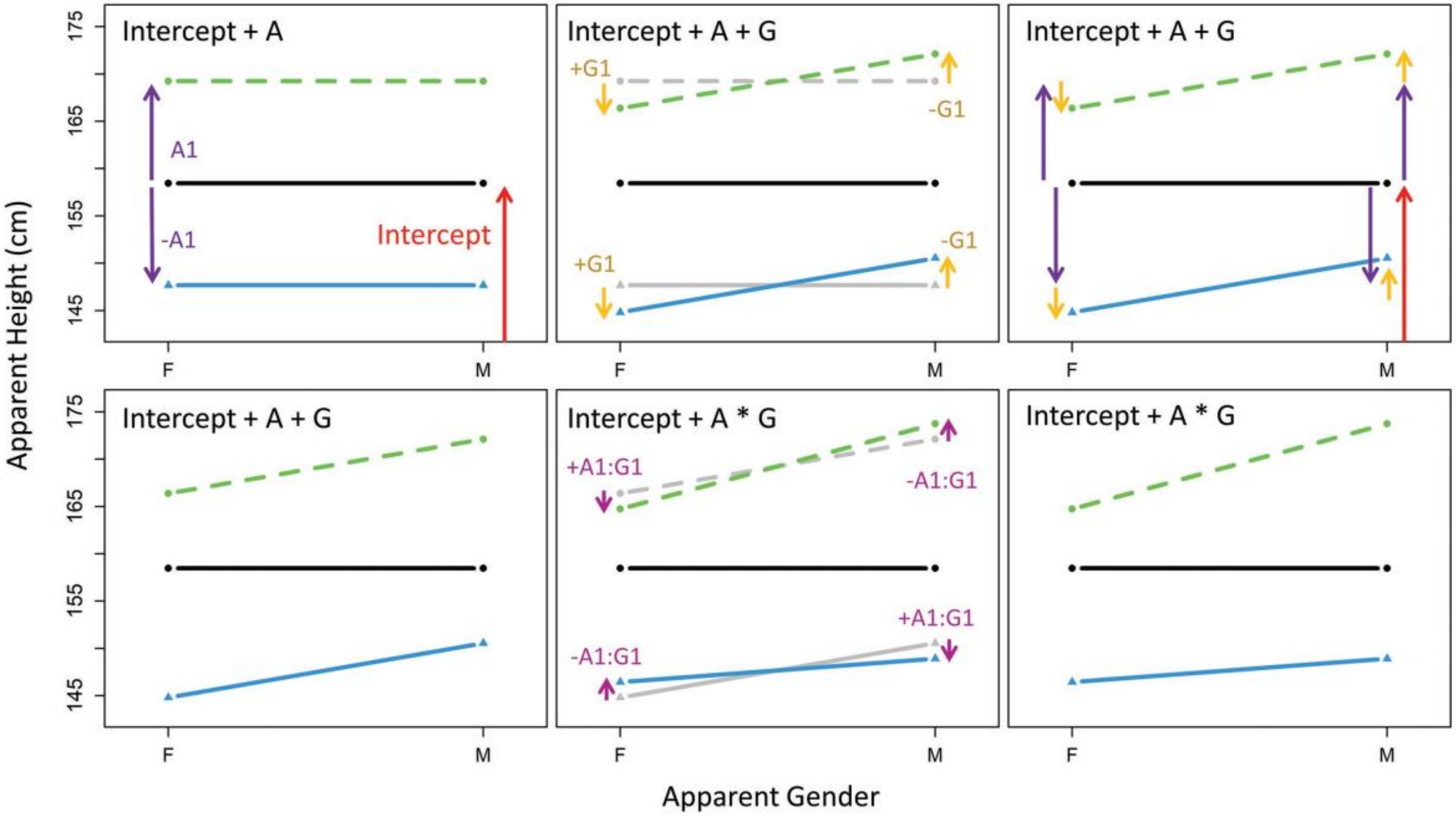
Fitting the Model

```
# Fit the model yourself
priors = c(brms::set_prior("student_t(3,156, 12)", class = "Intercept"),
           brms::set_prior("student_t(3,0, 12)", class = "b"),
           brms::set_prior("student_t(3,0, 12)", class = "sd"),
           brms::set_prior("lkj_corr_cholesky (2)", class = "cor"),
           brms::set_prior("gamma(2, 0.1)", class = "nu"),
           brms::set_prior("student_t(3,0, 12)", class = "sigma"))

model_interaction =
  brms::brm (height ~ A + G + A:G + (A + G + A:G|L) + (1|S),
             data = exp_data, chains = 4, cores = 4, warmup = 1000,
             iter = 5000, thin = 4, prior = priors, family = "student")

# Or download it from the GitHub page:
model_interaction = bmmb::get_model ('7_model_interaction.RDS')
```

```
bmmb::short_summary (model_interaction)
## Formula:  height ~ A + G + A:G + (A + G + A:G | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##                        Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)              4.24      0.79     2.96     6.03
## sd(A1)                     4.47      0.85     3.13     6.42
## sd(G1)                     2.10      0.49     1.37     3.22
## sd(A1:G1)                  1.34      0.36     0.76     2.13
## cor(Intercept,A1)         -0.71      0.13    -0.90    -0.38
## cor(Intercept,G1)         -0.20      0.22    -0.59     0.24
## cor(A1,G1)                -0.24      0.21    -0.62     0.20
## cor(Intercept,A1:G1)       0.17      0.23    -0.29     0.58
## cor(A1,A1:G1)             -0.02      0.23    -0.46     0.43
## cor(G1,A1:G1)             -0.34      0.24    -0.74     0.18
##
## ~S (Number of levels: 139)
##                 Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)       2.36      0.31     1.79     2.99
##
## Population-Level Effects:
##             Estimate Est.Error l-95% CI u-95% CI
## Intercept     158.46      1.12   156.22   160.62
## A1             10.78      1.21     8.39    13.15
## G1             -2.87      0.60    -4.07    -1.72
## A1:G1          -1.64      0.41    -2.44    -0.81
##
## Family Specific Parameters:
##         Estimate Est.Error l-95% CI u-95% CI
## sigma       5.01      0.16     4.70     5.34
## nu          3.44      0.33     2.87     4.15
```

Interpreting the Fixed Effects

```
## Population-Level Ef
##               Estimate
## Intercept     158.46
## A1             10.78
## G1             -2.87
## A1:G1          -1.64
##
```



Calculating the “Missing” Effects

- Si cambiamos entre positivo/negativo obtenemos el valor de parametros que no ajustamos directamente (los de rojo no ‘existen’).
  - -G1 = G2 el efecto de masculinidad aparente.
  - -A1:G1 = A1:G2 interacción entre adultez aparente y masculinidad.
  - -A1:G1 = A2:G1 interacción entre niñez aparente y femininidad.
  - --A1:G1 = A1:G1 = A2:G2 interacción entre la niñez aparente y masculinidad.

Recreating the Group Means

```
# intercept, boys, girls, men, women
means_pred_interaction = bmmb::short_hypothesis (
  model_interaction,
  c("Intercept + -A1 + -G1 +  A1:G1 = 0",  # boys
    "Intercept + -A1 +  G1 + -A1:G1 = 0",  # girl
    "Intercept +  A1 + -G1 + -A1:G1 = 0",  # men
    "Intercept +  A1 +  G1 +  A1:G1 = 0")) # women
```

```
# actual data means
tapply (exp_data$height, exp_data$C, mean)
##     b     g     m     w
## 146.9 145.0 174.5 165.6

# predictions with no interaction term
means_pred[,1]
## [1] 150.9 146.5 170.8 166.4

# predictions with interaction term
means_pred_interaction[,1]
## [1] 148.9 146.4 173.8 164.7
```

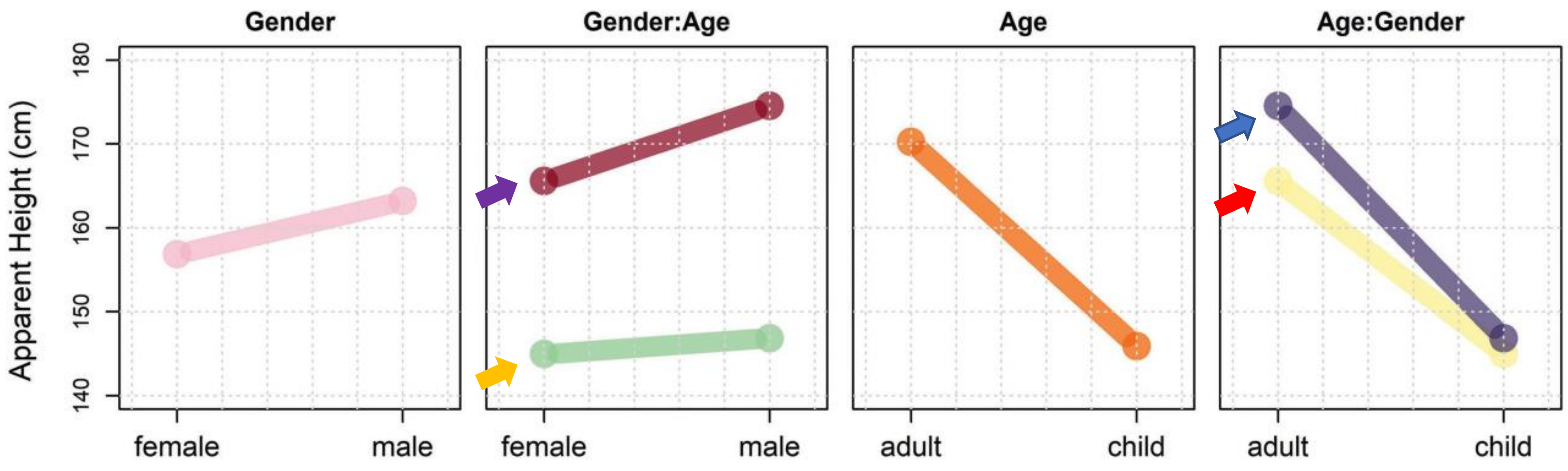
Calculating Simple Effects

- Los efectos simples se pueden calcular agregando efectos principales a las interacciones.
  - A1 = el efecto de adultez aparente
  - G1 = el efecto de feminidad aparente
  - A1:G1 = interacción entre adultez aparente y la feminidad.

- Por ejemplo:
  - A1 + A1:G1 = El efecto de adultez dada feminidad aparente
  - A1 - A1:G1 = El efecto de adultez dada masculinidad aparente
  - G1 + A1:G1 = El efecto de genero dada adultez aparente
  - G1 - A1:G1 = El efecto de genero dada niñez aparente

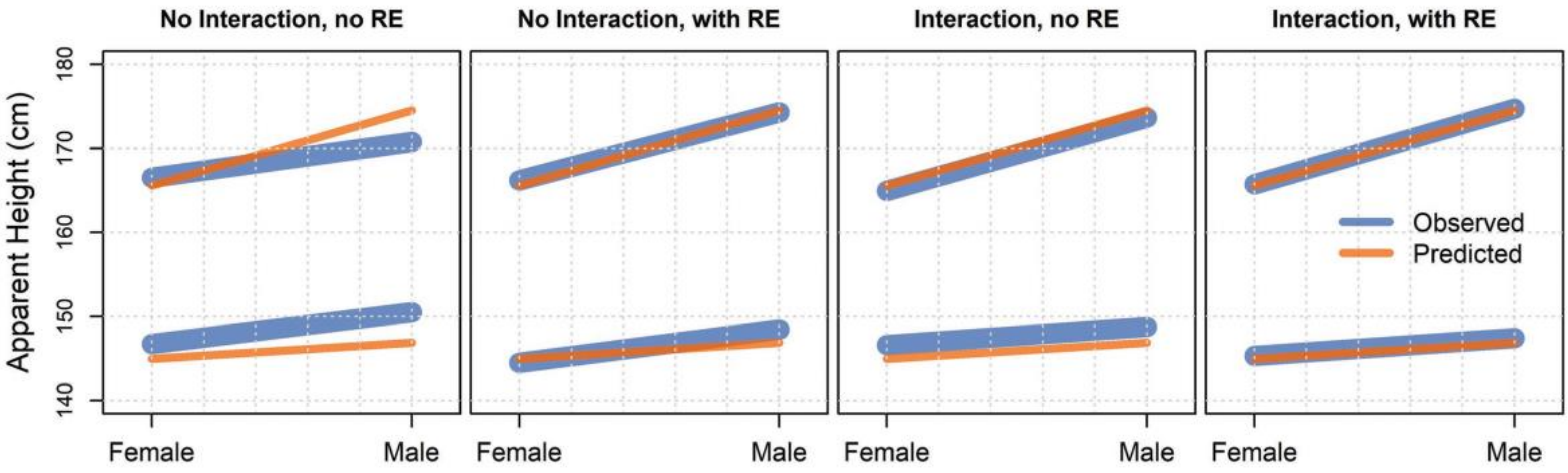
Calculating Simple Effects

```
# intercept, boys, girls, men, women
simple_effects = bmmb::short_hypothesis (
  model_interaction,
  c("A1 + A1:G1 = 0",  # effect for apparent age for females (G1)
    "A1 - A1:G1 = 0",  # effect for apparent age for males     (-G1)
    "G1 + A1:G1 = 0",  # effect for apparent gender for adults (A1)
    "G1 - A1:G1 = 0")) # effect for apparent gender for children (-A1)

# predictions with interaction term
simple_effects
##     Estimate Est.Error   Q2.5   Q97.5     hypothesis
## H1     9.142    1.2792  6.639 11.5977 (A1+A1:G1) = 0
## H2    12.424    1.2707  9.938 14.9913 (A1-A1:G1) = 0
## H3    -4.514    0.6174 -5.731
## H4    -1.232    0.8192 -2.860
```



Posterior Prediction

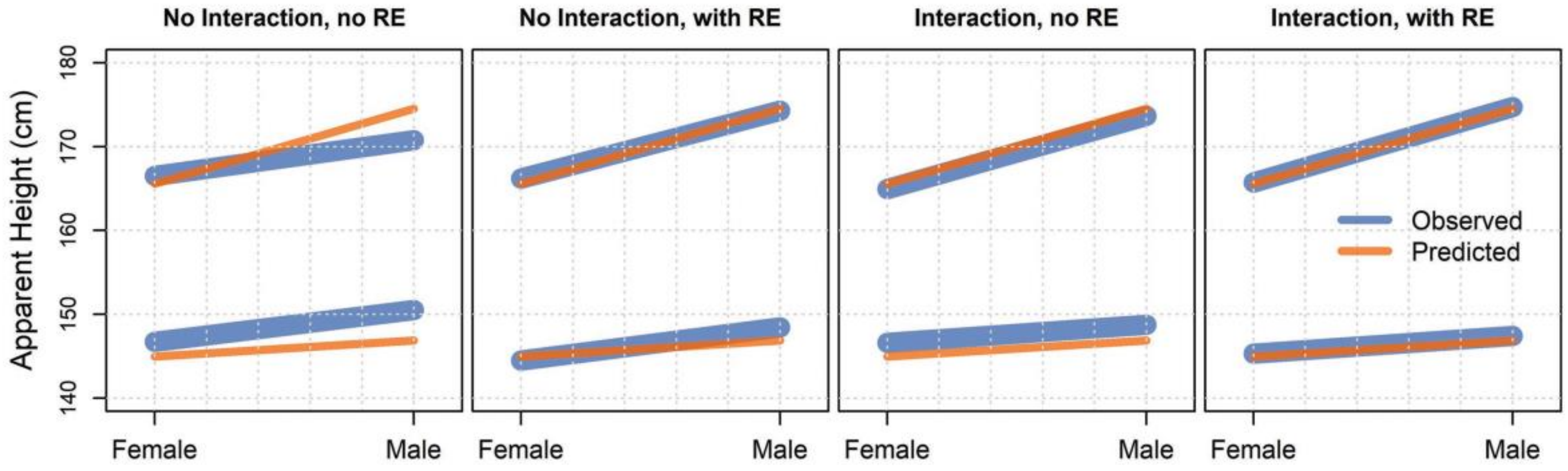
```
y_post_pred_int = predict (model_interaction)
y_post_pred_no_re_int = predict (model_interaction, re_formula = NA)
```



Model Comparison

```
model_both = brms::add_criterion (model_both, criterion="loo")
model_interaction = brms::add_criterion (model_interaction, criterion="loo")
```

```
brms::loo_compare (model_both, model_interaction)
##                      elpd_diff se_diff
## model_interaction     0.0        0.0
## model_both          -29.1       11.2
```



R^2 ('R-squared')

Descomposición de la varianza

$$\sigma^2_{\text{total}} = \sigma^2_{\text{explained}} + \sigma^2_{\text{error}}$$

Relación entre la varianza explicada y el total

$$R^2 = \frac{\sigma^2_{\text{explained}}}{\sigma^2_{\text{total}}} = \frac{\sigma^2_{\text{explained}}}{\sigma^2_{\text{explained}} + \sigma^2_{\text{error}}}$$

Equivalente bayesiano, para la muestra posterior S.

$$R^2_s = \frac{V_{i=1}^{n}\ \hat{y}_i^s}{V_{i=1}^{n}\ \hat{y}_i^s + V_{i=1}^{n}\ \hat{e}_i^s}$$

$$\hat{e}_i^s = \hat{y}_i^s - y_i$$

residuals

predicted data

observed data

R^2 Comparison

```
r2_both = r2_bayes(model_both)
r2_interaction = r2_bayes(model_interaction)
```

```
r2_both
##         Estimate Est.Error   Q2.5  Q97.5
## [1,]      0.7755  0.005566 0.7641 0.7858

r2_interaction
##         Estimate Est.Error   Q2.5  Q97.5
## [1,]      0.7802  0.005347 0.7694 0.7901
```

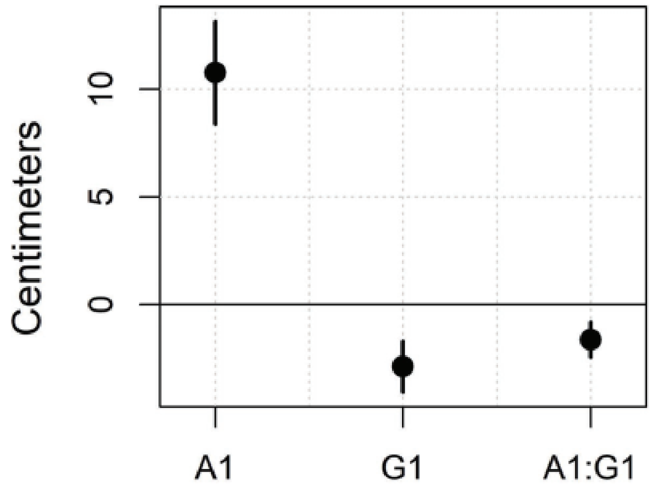
```
r2_both_no_re = r2_bayes(model_both, re_formula = NA)
r2_interaction_no_re = r2_bayes(model_interaction, re_formula = NA)
```

```
r2_both_no_re
##         Estimate Est.Error   Q2.5  Q97.5
## [1,]       0.503   0.07258 0.3411 0.6216
```

```
r2_interaction_no_re
##         Estimate Est.Error   Q2.5  Q97.5
## [1,]      0.5732   0.05909 0.4389 0.6678
```

Answering Our Research Questions

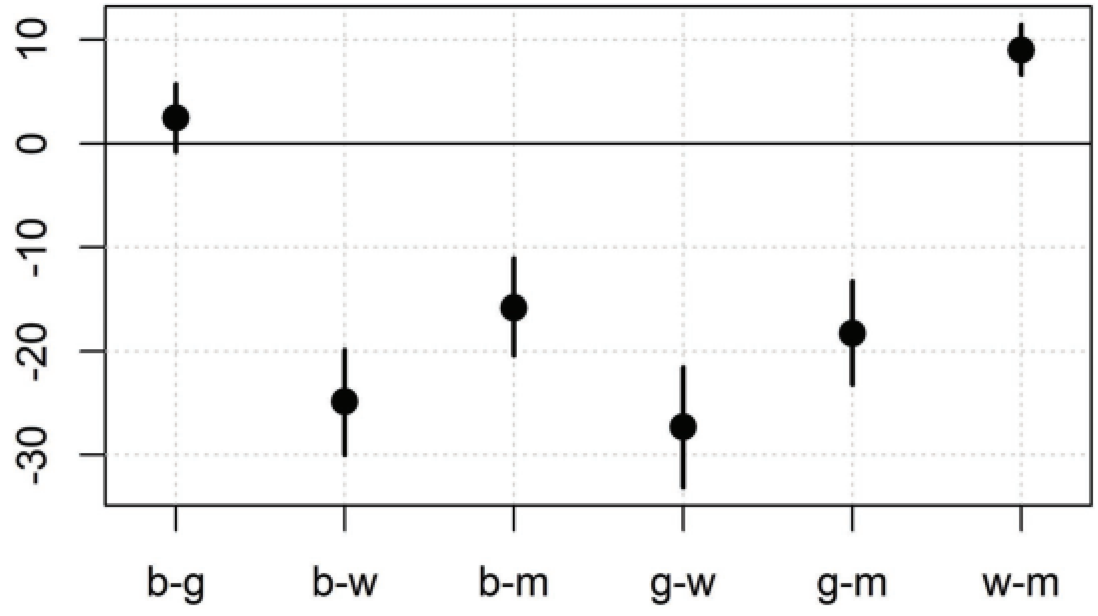
Results indicate that the average apparent height across all speaker groups (i.e. the intercept) was 158.5 (s.d. = 1.12, 95% C.I = [156.22, 160.62]). We also found an average effect of 10.8 cm for apparent speaker age (s.d. = 1.21, 95% C.I = [8.39, 13.15]) and −2.9 cm for apparent speaker gender (s.d. = 0.6, 95% C.I = [−4.07, −1.72]). In addition, we found an interaction between the effects of apparent age and apparent gender on apparent heights (mean = −1.64, s.d. = 0.41, 95% C.I = [−2.44, -0.81]). The result of these effects is that apparent adults were perceived as taller than apparent children and apparent males were perceived as taller than apparent females. However, the difference in apparent height due to apparent gender was larger for adults than for children (and the effect for apparent age was larger for males than for females). Figure 7.11 presents the model's fixed effects other than the intercept (whose value is too large to plot in this range).



Pairwise Group Differences

```r
# get group mean predictions
C = attributes(means_pred_interaction)$samples

# find pairwise differences
pairwise_diffs = cbind("b-g"=C[,1]-C[,2], "b-w"=C[,1]-C[,3],
                       "b-m"=C[,1]-C[,4], "g-w"=C[,2]-C[,3],
                       "g-m"=C[,2]-C[,4], "w-m"=C[,3]-C[,4])

# summarize these
pairwise_diffs_summary = posterior_summary(pairwise_diffs)
```

¿Cuáles son 'reales'?

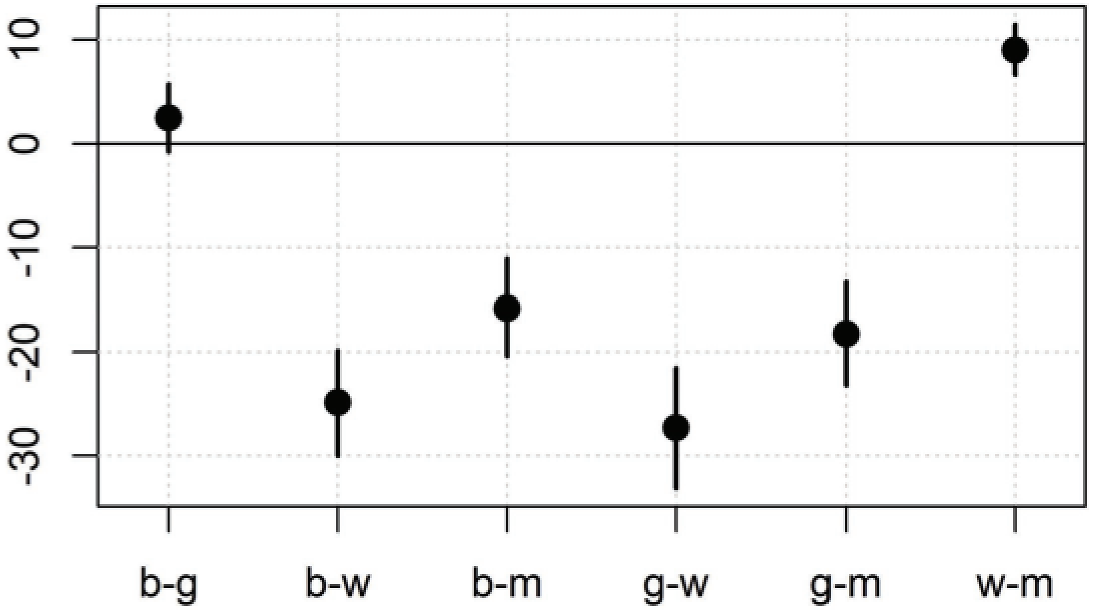


Is an Effect 'Real'?

- La mayoría de los intervalos de credibilidad del 95% no cruzan 0. ¿Esto hace que los efectos sean "reales"?

- Los intervalos creíbles b-g cruzan 0. ¿Significa esto que el efecto es igual a 0?

- Incluso si un intervalo cruza cero, su valor más probable no suele ser cero.

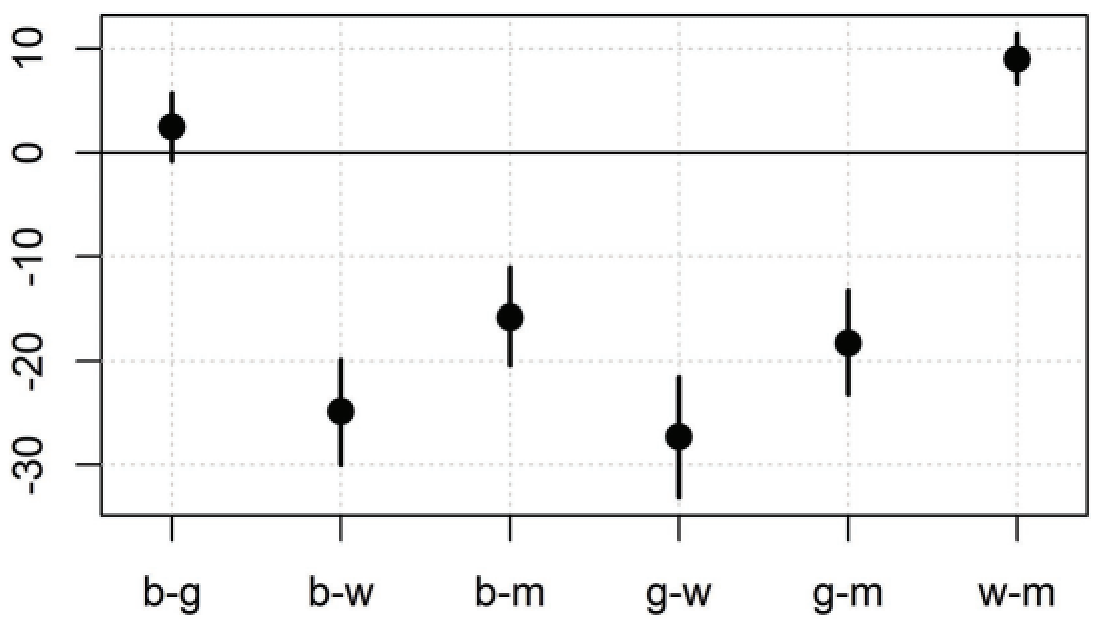


Type S and M Errors

- Mejor manera de pensarlo: Evite los errores de tipo S y M.

- Errores de tipo S (signo): ¿Es el valor negativo cuando cree que es positivo (o viceversa)?

- Errores de tipo M (magnitud): ¿Es el valor pequeño cuando cree que es grande (o viceversa)?

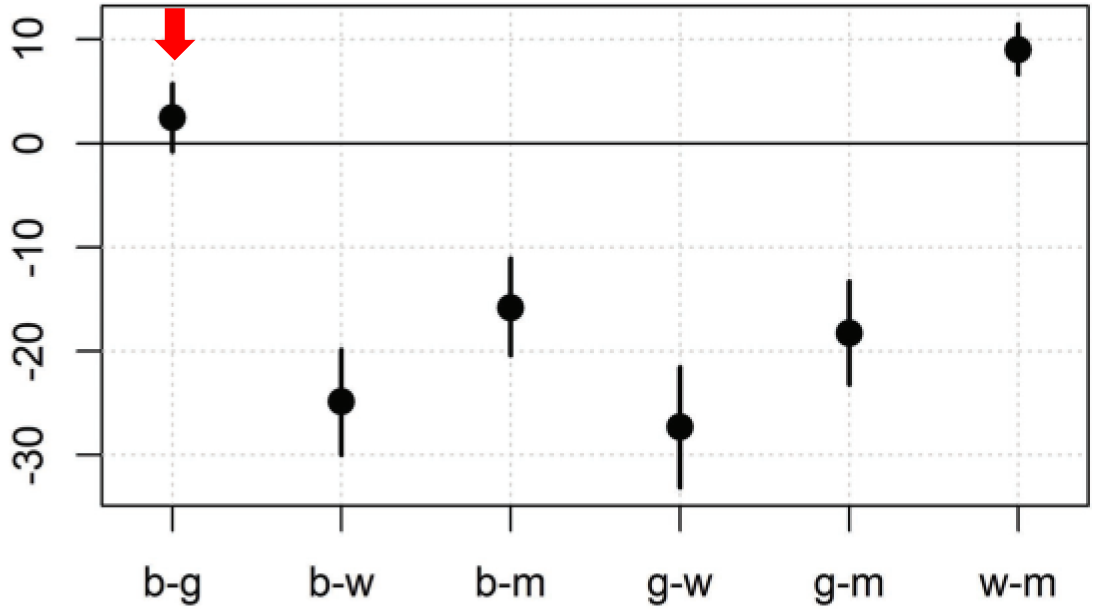


Region of Practical Equivalence

- Región de Equivalencia Práctica (ROPE): ¿Qué tan pequeño tiene que ser un efecto antes de que un efecto pueda ser cero?

- Ejemplo: Para la altura humana, las diferencias inferiores a 0.5 cm tienen poco significado práctico. Una diferencia "significativa" de 0.01 cm bien podría ser cero.

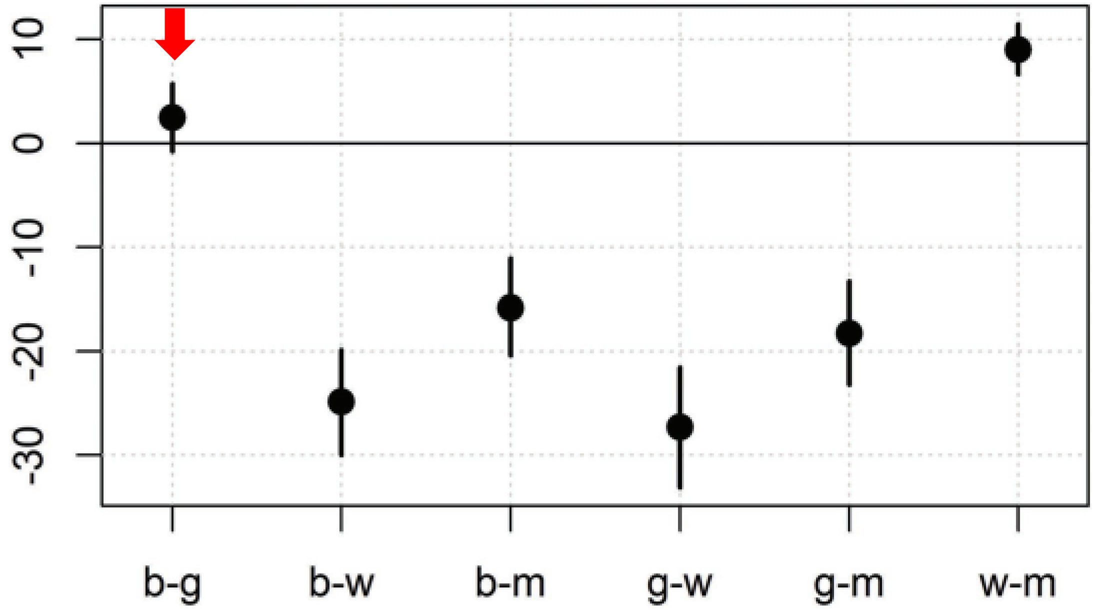
```
# summary of difference between boy and girl means
pairwise_diffs_summary[1,]
##   Estimate Est.Error      Q2.5     Q97.5
##     2.4643    1.6384   -0.7837    5.7199
```



So, is the b-g Difference 'Real'?

- Se trata de una afirmación epistemológica u ontológica.
- Las estadísticas pueden ayudarte a hacer estas afirmaciones, pero tu eres responsable de ellas.
- Conclusión: La diferencia puede ser real y positiva, pero probablemente sea pequeña. Es necesario realizar más investigaciones.

```
# summary of difference between boy and girl means
pairwise_diffs_summary[1,]
##   Estimate Est.Error       Q2.5      Q97.5
##     2.4643    1.6384    -0.7837     5.7199
```



Factors with more Levels

$$\begin{bmatrix} A1 & A2 & A3 & (A4) \end{bmatrix}$$

$$\begin{bmatrix} B1 & B2 & B3 & (B4) \end{bmatrix}$$

$$\begin{bmatrix} A1 & A2 & A3 & -(A1+A2+A3) \end{bmatrix}$$

$$\begin{bmatrix} B1 & B2 & B3 & -(B1+B2+B3) \end{bmatrix}$$

Factors with more Levels

$$\begin{bmatrix} A1:B1 & A2:B1 & A3:B1 & (A4:B1) \\ A1:B2 & A2:B2 & A3:B2 & (A4:B2) \\ A1:B3 & A2:B3 & A3:B3 & (A4:B3) \\ (A1:B4) & (A2:B4) & (A3:B4) & (A4:B4) \end{bmatrix}$$

$$\begin{bmatrix} A1:B1 & A2:B1 & A3:B1 & -(A1:B1+A2:B1+A3:B1) \\ A1:B2 & A2:B2 & A3:B2 & (A4:B2) \\ A1:B3 & A2:B3 & A3:B3 & (A4:B3) \\ -(A1:B1+A1:B2+A1:B3) & (A2:B4) & (A3:B4) & (A4:B4) \end{bmatrix}$$

Exercises

Use the data in ‘exp_ex’ to do one of the following. You may also use your own data to answer a related question. In any case, describe the model, present and explain the results, and include at least two figures.

1. Medium: Fit and interpret the pre-fit model (or a similar model that you fit). Report the results of the model. Recreate the predicted group means and report and interpret at least one simple effect.

2. Hard: Fit and interpret some other model that includes at least two factors with at least two levels each, and their interactions. Report the results of the model. Recreate the predicted group means and report and interpret at least one simple effect.